

G. Lejeune Dirichlet's Werke. II.

III.

**On
the Characteristic Properties
of the Potential of a Mass Distributed
over One or More Finite Surfaces.**

By G. Lejeune Dirichlet.

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**On
the Characteristic Properties of the Potential
of a Mass Distributed over One or More Finite Surfaces.**

As was already observed in an earlier paper,¹ the method developed there for the case of a mass extended in three dimensions can also be applied to the question just mentioned, and it follows that, for a mass distributed over surfaces, the characteristic properties of the potential, that is, those which completely determine its expression, are the following:

1) The potential v is, throughout all of space, a finite and continuous function of the rectangular coordinates x, y, z .

2) The three partial differential quotients of the potential taken with respect to the coordinates are likewise continuous everywhere outside the surfaces. For a point lying on them, however, there is an interruption of continuity consisting in this: if the potential at such a point and at the points lying on the normal on both sides at distance ε are denoted by v, v', v'' , then the quotient

$$\frac{v' + v'' - 2v}{\varepsilon}$$

has, for an infinitely small positive ε , the expression $-4\pi\rho$ as its limit, where ρ denotes the density at the point of the surface.

3) Everywhere outside the surfaces the equation

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$$

holds.

4) At infinite distance from the surfaces, the same conditions are fulfilled as for a mass extended in three dimensions.

Apart from the utility that may be afforded by the proof that the properties just listed completely characterize the potential, in order to verify an expression for the potential known from wherever it may have been obtained, this proof also presents another and more essential interest. It follows from it, namely, that the important theorem stated by Gauss, according to which a mass can always be distributed over given surfaces in such a way that the potential corresponding to this distribution receives, at every point of the surfaces, an arbitrarily prescribed value varying according to continuity, is wholly identical with the assertion that in a homogeneous mass filling all of space, provided that initially only temperatures vanishing at infinite distance

¹*Crelle's Journal*, Band 32, p. 80. Editorial note in the Werke: P. 9 of the second volume of this edition of G. Lejeune Dirichlet's works. The observation is found at the end of the paper on p. 16. K.

occur, a fixed temperature will always be established everywhere after an infinite time under the influence of constant heat sources distributed over surfaces. The identity of the two theorems, and the new relation between two branches of mathematical physics already exhibiting such great kinship that is thereby established, is immediately clear from what has been said above, since the known conditions determining the permanent state of heat coincide exactly with those by which the potential corresponding to the required mass distribution is characterized.